

CN LAB 3

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LAB 3.1

The screenshot shows a Linux desktop with a blue-themed background. A terminal window is open, displaying the command `dig www.google.com` and its output. Below the terminal, a Wireshark packet capture window is open, showing a list of captured packets. The first packet is a DNS Standard query from 10.0.2.15 to 10.0.2.2. The second packet is a DNS Standard query response from 10.0.2.2 to 10.0.2.15. The third packet is a Router Solicitation from 10.0.2.15 to 10.0.2.2. The fourth packet is a DNS Standard query response from 10.0.2.2 to 10.0.2.15. The fifth packet is a CNAME record for ncredir-lb.wikimedia.org. The sixth packet is a DHCP Request from 10.0.2.15 to 10.0.2.2. The seventh packet is a DHCP ACK from 10.0.2.2 to 10.0.2.15. The eighth packet is a Who has 10.0.2.2? Tell 10.0.2.15 from 10.0.2.15 to 10.0.2.2. The ninth packet is a Multicast Listener Report Message v2 from 10.0.2.15 to 10.0.2.2. The tenth packet is a Destination unreachable (Port unreachable) from 10.0.2.2 to 10.0.2.15.

```
dhriti@192:~  
$ dig www.google.com  
  
;<<> Dig 9.20.15-2-Debian <<> www.google.com  
;; global options: +cmd  
;; Got answer:  
;;->HEADER-<- opcode: QUERY, status: NOERROR, id: 20962  
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 0  
  
;; QUESTION SECTION:  
;; www.google.com.  
IN A  
  
;; ANSWER SECTION:  
www.google.com. 149 IN A 142.250.206.36  
  
;; Query time: 16 msec  
;; SERVER: 192.168.254.1#53(192.168.254.1) (UDP)  
;; WHEN: Thu Feb 12 14:52:09 IST 2026  
;; MSG SIZE rcvd: 48
```

Protocol Length Info
DNS 97 Standard query 0x51e2 A www.google.com OPT
DNS 90 Standard query response 0x51e2 A www.google.com A 142.250.206.36
ICMPv6 62 Router Solicitation
DNS 100 Standard query 0xc6df A www.wikipedia.com OPT
DNS 131 Standard query response 0xc6df A www.wikipedia.com CNAME ncredir-lb.wikimedia.org
ARP 42 Who has 10.0.2.2? Tell 10.0.2.15
ARP 60 10.0.2.2 is at 52:54:00:12:35:02
ICMPv6 90 Multicast Listener Report Message v2
DHCP 329 DHCP Request - Transaction ID 0x22d6b85e
DHCP 590 DHCP ACK - Transaction ID 0x22d6b85e
ARP 42 Who has 10.0.2.2? Tell 10.0.2.15
ARP 60 10.0.2.2 is at 52:54:00:12:35:02
ICMP 590 Destination unreachable (Port unreachable)

Frame 1: Packet, 97 bytes on wire (776 bits), 97 bytes captured (776 bits) on eth0
Ethernet II, Src: PCSSystemtec_bc:43:40 (08:00:27:bc:43:40), Dst: 08:00:27:fe:01:ba
Internet Protocol Version 4, Src: 10.0.2.15, Dst: 10.0.2.2
User Datagram Protocol, Src Port: 47721, Dst Port: 53
Domain Name System (query)

The screenshot shows a Linux desktop with a blue-themed background. A terminal window is open, displaying the command `dig www.wikipedia.com` and its output. Below the terminal, a Wireshark packet capture window is open, showing a list of captured packets. The first packet is a DNS Standard query from 10.0.2.15 to 10.0.2.2. The second packet is a DNS Standard query response from 10.0.2.2 to 10.0.2.15. The third packet is a Router Solicitation from 10.0.2.15 to 10.0.2.2. The fourth packet is a DNS Standard query response from 10.0.2.2 to 10.0.2.15. The fifth packet is a CNAME record for ncredir-lb.wikimedia.org. The sixth packet is a DHCP Request from 10.0.2.15 to 10.0.2.2. The seventh packet is a DHCP ACK from 10.0.2.2 to 10.0.2.15. The eighth packet is a Who has 10.0.2.2? Tell 10.0.2.15 from 10.0.2.15 to 10.0.2.2. The ninth packet is a Multicast Listener Report Message v2 from 10.0.2.15 to 10.0.2.2. The tenth packet is a Destination unreachable (Port unreachable) from 10.0.2.2 to 10.0.2.15.

```
dhriti@192:~  
$ dig www.wikipedia.com  
  
;<<> Dig 9.20.15-2-Debian <<> www.wikipedia.com  
;; global options: +cmd  
;; Got answer:  
;;->HEADER-<- opcode: QUERY, status: NOERROR, id: 50911  
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 0  
  
;; QUESTION SECTION:  
;; www.wikipedia.com.  
IN A  
  
;; ANSWER SECTION:  
www.wikipedia.com. 21145 IN CNAME ncredir-lb.wikimedia.org.  
ncredir-lb.wikimedia.org. 180 IN A 103.102.166.226  
  
;; Query time: 188 msec  
;; SERVER: 192.168.254.1#53(192.168.254.1) (UDP)  
;; WHEN: Thu Feb 12 14:55:26 IST 2026  
;; MSG SIZE rcvd: 89
```

Protocol Length Info
DNS 97 Standard query 0x51e2 A www.google.com OPT
DNS 90 Standard query response 0x51e2 A www.google.com A 142.250.206.36
ICMPv6 62 Router Solicitation
DNS 100 Standard query 0xc6df A www.wikipedia.com OPT
DNS 131 Standard query response 0xc6df A www.wikipedia.com CNAME ncredir-lb.wikimedia.org
ARP 42 Who has 10.0.2.2? Tell 10.0.2.15
ARP 60 10.0.2.2 is at 52:54:00:12:35:02
ICMPv6 90 Multicast Listener Report Message v2
DHCP 329 DHCP Request - Transaction ID 0x22d6b85e
DHCP 590 DHCP ACK - Transaction ID 0x22d6b85e
ARP 42 Who has 10.0.2.2? Tell 10.0.2.15
ARP 60 10.0.2.2 is at 52:54:00:12:35:02
ICMP 590 Destination unreachable (Port unreachable)

Frame 4: Packet, 100 bytes on wire (800 bits), 100 bytes captured (800 bits) on eth0
Ethernet II, Src: PCSSystemtec_bc:43:40 (08:00:27:bc:43:40), Dst: 08:00:27:fe:01:dc
Internet Protocol Version 4, Src: 10.0.2.15, Dst: 10.0.2.2
User Datagram Protocol, Src Port: 56463, Dst Port: 53
Domain Name System (query)

1. How does UDP differ from TCP?

UDP is connectionless and does not use a handshake before sending data.

TCP is connection-oriented and uses a three-way handshake.

UDP is faster but does not guarantee delivery.

TCP is slower but reliable and ensures data is delivered correctly.

2. Why is UDP suitable for DNS?

DNS queries are small and need fast responses.

UDP is lightweight and does not require connection setup.

This makes DNS resolution quick and efficient.

DNS uses UDP on port 53.

3. Observations about UDP traffic in Wireshark

UDP packets were seen between two IP addresses.

Both systems used port 12345.

No handshake was observed.

Messages were sent instantly.

Data was visible inside the UDP packets.

LAB 3.2

The screenshot displays a web-based UDP Chat application interface. It is divided into several sections:

- UDP Settings:** Located at the top left, it includes a text input for "Friend's IP:" with the value "10.131.93.44", a "Port:" dropdown set to "12345", and a "Start Chat" button.
- Chat:** A large text area on the left showing the chat history. The messages are:
 - "UDP Chat initialized on port 12345. You can start sending messages now."
 - "No connection required - UDP is connectionless!"
 - "[15:23:02] 10.131.93.44: hi!"
 - "[15:23:05] You: heyyyyyyyyyy"
 - "[15:23:08] You: wassssuppppp"
 - "[15:23:09] 10.131.93.44: aaaaaayyyyyyy"
- Network Statistics:** A panel on the right showing session details:
 - Status: Chatting (UDP)
 - Local IP: 10.131.93.35
 - Port: 12345
 - Messages Sent: 3
 - Messages Received: 3
 - Bytes Sent: 42 bytes
 - Bytes Received: 42 bytes
 - Network Interface: Wi-Fi
 - Session Duration: 200 seconds
- Network Concepts:** A scrollable list of concepts on the right side:
 - ◊ UDP (User Datagram Protocol):
 - A connectionless protocol
 - Faster than TCP but less reliable
 - No guaranteed delivery or order
 - Perfect for real-time applications
 - ◊ Socket Programming:
 - Enables network communication
 - Uses IP addresses and ports
 - Allows data exchange between computers
 - ◊ IPv4 Addressing:
 - Format: xxx.xxx.xxx.xxx
 - Each number: 0-255
 - Used to identify devices on network
- Send:** A text input field at the bottom left with a "Send" button.

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

udp.port == 12345

No.	Time	Source	Destination	Protocol	Length	Info
1088	16.129263	192.168.1.35	192.168.1.37	UDP	49	12345 → 12345 Len=7
1560	22.632574	192.168.1.37	192.168.1.35	UDP	52	12345 → 12345 Len=10
2196	31.806530	192.168.1.35	192.168.1.37	UDP	69	12345 → 12345 Len=27
2616	35.624444	192.168.1.37	192.168.1.35	UDP	51	12345 → 12345 Len=9
2816	37.473676	192.168.1.37	192.168.1.35	UDP	50	12345 → 12345 Len=8
4243	48.351013	192.168.1.35	192.168.1.37	UDP	55	12345 → 12345 Len=13

> Frame 1088: Packet, 49 bytes on wire (392 bits)

> Ethernet II, Src: Intel_13:05:33 (e4:1f:d5:13:05:33), Dst: 08:00:00:08:00:00

> Internet Protocol Version 4, Src: 192.168.1.35, Dst: 192.168.1.37

> User Datagram Protocol, Src Port: 12345, Dst Port: 12345

Source Port: 12345

Destination Port: 12345

Length: 15

Checksum: 0x83b9 [unverified]

[Checksum Status: Unverified]

[Stream index: 8]

[Stream Packet Number: 1]

[Timestamp]

0000 ac d5 64 cf 67 c3 e4 1f d5 13 05 33 08 00 45

0010 00 23 c2 fe 00 00 80 11 00 00 c0 a8 01 23 c0

0020 01 25 30 39 30 39 00 0f 83 b9 68 65 79 79 79

0030 79

wireshark_Wi-FiFOLLK3.pcapng

Packets: 7089 · Displayed: 6 (0.1%) · Dropped: 0 (0.0%)

Profile: Default

LAB 3.3

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1083	16.090421	192.168.1.35	162.128.44.82	DNS	217	Unknown operation
1084	16.107876	192.168.1.35	162.128.44.82	DNS	211	Unknown operation
1085	16.110422	162.128.44.82	192.168.1.35	DNS	324	Unknown operation
1086	16.119881	162.128.44.82	192.168.1.35	DNS	331	Unknown operation
1087	16.128117	192.168.1.35	162.128.44.82	DNS	207	Unknown operation
1088	16.129263	192.168.1.35	192.168.1.37	UDP	49	12345 → 12345 Len=7
1089	16.137143	162.128.44.82	192.168.1.35	DNS	325	Unknown operation
1090	16.147921	192.168.1.35	162.128.44.82	DNS	206	Inverse query response
1091	16.152150	162.128.44.82	192.168.1.35	DNS	327	Dynamic update 0x1...
1092	16.169926	192.168.1.35	162.128.44.82	DNS	206	Unknown operation

> Frame 1088: Packet, 49 bytes on wire (392 bits)

> Ethernet II, Src: Intel_13:05:33 (e4:1f:d5:13:05:33), Dst: 08:00:00:08:00:00

> Internet Protocol Version 4, Src: 192.168.1.35, Dst: 192.168.1.37

> User Datagram Protocol, Src Port: 12345, Dst Port: 12345

Source Port: 12345

Destination Port: 12345

Length: 15

Checksum: 0x83b9 [unverified]

[Checksum Status: Unverified]

[Stream index: 8]

[Stream Packet Number: 1]

[Timestamp]

0000 ac d5 64 cf 67 c3 e4 1f d5 13 05 33 08 00 45

0010 00 23 c2 fe 00 00 80 11 00 00 c0 a8 01 23 c0

0020 01 25 30 39 30 39 00 0f 83 b9 68 65 79 79 79

0030 79

wireshark_Wi-FiFOLLK3.pcapng

Packets: 7089 · Dropped: 0 (0.0%)

Profile: Default